Analytical Chemistry Cumulative Examination
December 7, 2001

Please answer the following questions about pH, acid-base equilibria, and titrations. All work must be shown. Be sure to indicate the logic used to solve each problem.

1. Define pH and pKa. Also, give a *general* equilibrium expression for Ka in terms of the concentrations of different chemical species involved in the equilibrium of HA (a monoprotic acid) with water. (15 points)

2. pH is most commonly measured with a glass electrode. Draw a schematic diagram of a glass pH electrode, identify all parts, and explain their purposes. (20 points)

3. Suppose that a solution of sodium hydroxide is permitted to absorb large amounts of carbon dioxide after originally being standardized. Would the original value for the concentration of the sodium hydroxide titrant be high, low, or correct? Explain. (10 points)

4. Determine the pH *at equilibrium* of an aqueous solution that has analytical concentrations of 0.050 M oxalic acid (H_2A) and 0.025 M sodium oxalate (NaHA). (K_{a1}= 5.26 x 10^{-2}, K_{a2}= 5.42 x 10^{-5}). (15 points)

5. Calculate the pH of the solution when 50.00 mL of 0.05 M acetic acid (HA, K_{a1}= 1.75 x 10^{-5}) is titrated with the following volumes of 0.100 M NaOH:

   a) 0.00 mL NaOH (5 points)

   b) 1.00 mL NaOH (5 points)

   c) 25.00 mL NaOH (5 points)

   d) 25.40 mL NaOH (5 points)

6. A hydrochloric acid solution was standardized against sodium carbonate. Just 37.66 mL of the acid was required to titrate a 0.3663 g sample of the sodium carbonate. Later it was discovered that the sodium carbonate was really the hydrated compound, $Na_2CO_3 \cdot 10\ H_2O$, and not the anhydrous compound as expected. What were the reported and corrected concentrations of the hydrochloric acid solution? (20 points)

Relevant Information:

Atomic weights Na=22.99; C=12.01; O=16; H=1.01